

MARION, OH, USA

WMO: 725208

Lat: 40.616N

Lon: 83.064W

Elev: 303

StdP: 97.74

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04855

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.4	-21.3	0.6	-16.7	-18.6	0.8	-13.9	11.9	0.6	11.1	0.2	3.0	270	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.1	22.9	30.8	22.5	29.2	21.5	24.6	29.6	23.7	28.4	22.9	27.3	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	27.2	22.3	17.6	26.5	21.4	16.7	25.6	75.7	29.9	72.3	28.5	69.1	27.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.1	8.3	-21.2	33.5	4.0	1.6	-24.0	34.6	-26.4	35.5	-28.6	36.4	-31.5	37.5
			-21.5	26.0	3.8	1.1	-24.3	26.8	-26.5	27.4	-28.7	28.1	-31.5	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-3.2	-1.5	3.7	10.5	16.5	21.1	22.8	21.8	18.3	11.9	5.5	-0.2	(6)
	DBStd	10.47	6.86	6.44	6.16	5.23	4.66	3.45	2.95	2.78	4.03	5.07	5.31	5.84	(7)
	HDD10.0	1524	411	327	214	57	4	0	0	0	1	38	154	319	(8)
	HDD18.3	3249	668	557	454	241	92	14	2	4	49	208	386	574	(9)
	CDD10.0	1765	1	3	18	70	207	334	397	365	251	97	19	3	(10)
	CDD18.3	448	0	0	1	4	36	98	141	111	49	9	0	0	(11)
	CDH23.3	3811	0	0	5	52	320	812	1227	888	437	69	1	0	(12)
	CDH26.7	1134	0	0	1	4	76	253	413	245	130	12	0	0	(13)
(14) Wind	WSAvg	3.9	4.7	4.6	4.6	4.7	4.0	3.5	2.9	2.7	3.0	3.8	4.3	4.4	(14)
(15) Precipitation	PrecAvg	947	74	52	66	92	105	111	107	94	81	70	72	72	(15)
	PrecMax	1102	170	145	129	142	228	205	275	234	154	145	150	219	(16)
	PrecMin	646	27	19	31	40	43	34	38	36	27	16	17	23	(17)
	PrecStd	128	36	28	26	27	48	49	53	40	35	33	31	42	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	18.8	23.0	27.4	30.7	32.8	33.8	32.5	32.4	28.4	22.0	16.4	(19)
		MCWB	13.4	13.9	16.0	18.1	21.4	22.2	24.3	23.5	21.6	20.5	14.4	13.4	(20)
	2%	DB	12.7	14.0	19.2	24.9	28.8	31.2	32.1	31.1	30.2	25.1	18.3	13.2	(21)
		MCWB	11.0	11.2	13.9	16.2	19.7	22.1	23.7	23.0	21.4	17.9	13.7	11.2	(22)
	5%	DB	9.4	11.1	16.4	22.5	27.1	29.6	30.7	29.2	27.9	22.8	16.3	10.9	(23)
		MCWB	7.7	8.6	12.2	15.0	19.2	21.3	23.0	22.1	20.4	16.7	12.5	9.1	(24)
	10%	DB	6.4	7.5	13.7	19.9	24.9	27.8	29.0	27.8	26.0	20.4	13.9	8.0	(25)
		MCWB	4.3	4.9	10.1	13.5	18.1	20.5	22.0	21.5	19.0	15.8	10.8	6.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	14.5	16.9	19.8	22.7	24.2	25.8	25.5	24.0	21.8	16.9	13.8	(27)
		MCDB	15.0	17.7	21.8	25.1	27.8	29.8	31.2	30.0	29.4	26.9	19.7	15.7	(28)
	2%	WB	11.3	11.6	14.8	17.5	21.5	23.3	24.9	24.2	22.8	19.8	14.8	11.4	(29)
		MCDB	12.5	13.4	18.5	22.4	26.9	28.9	30.4	28.7	27.7	23.1	17.5	13.0	(30)
	5%	WB	8.0	8.4	12.8	15.9	20.2	22.5	24.0	23.4	21.6	18.0	13.0	9.1	(31)
		MCDB	9.2	10.8	15.8	20.7	25.0	27.8	28.7	27.4	25.9	21.3	15.5	10.4	(32)
	10%	WB	4.6	5.1	10.5	14.5	19.1	21.6	23.1	22.5	20.5	16.2	11.1	6.5	(33)
		MCDB	6.3	7.3	13.4	18.9	23.5	26.4	27.2	26.3	24.2	19.9	13.9	8.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	9.7	11.8	11.2	10.6	10.7	11.0	11.8	10.8	9.3	7.2	(35)
		MCDBR	10.0	12.2	13.2	14.7	13.4	12.7	11.7	12.1	13.7	13.5	12.4	10.0	(36)
	5% WB	MCWBR	9.0	9.7	8.8	8.3	6.6	5.4	4.9	5.6	6.3	7.3	8.8	8.9	(37)
		MCWBR	9.6	11.3	11.7	12.6	11.0	10.5	10.4	10.2	11.2	11.1	10.6	9.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.306	0.328	0.354	0.397	0.431	0.448	0.454	0.443	0.389	0.360	0.324	0.303	(40)
		taud	2.403	2.330	2.333	2.236	2.182	2.176	2.175	2.189	2.321	2.388	2.449	2.445	(41)
	Ebn at Noon		851	881	895	875	850	833	823	822	847	834	823	826	(42)
		Edh at Noon	82	102	114	135	145	147	146	140	114	96	78	73	(43)
(44) All-Sky Solar Radiation	RadAvg		1.56	2.46	3.44	4.65	5.48	6.02	5.97	5.40	4.37	2.95	1.94	1.32	(44)
	RadStd		0.15	0.30	0.33	0.41	0.46	0.45	0.35	0.23	0.37	0.25	0.27	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+61

Nomenclature: See separate page